RAFFLES JUNIOR COLLEGE

Preliminary Examination

H2 BIOLOGY **9747/2**

PAPER 2 **9th September 2008**

2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.
Write your answers **ONLY in the spaces** provided on the question paper.

Section B

Answer any **ONE** question.
Write your answer on the separate answer paper provided.
All working for numerical answers must be shown.

At the end of the examination, **hand in your essay SEPARATELY.**

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
Total	

This document consists of **15** printed pages.

Raffles Junior College
Internal Examination

[Turn over

Section A

Answer **all** the questions in this section.

1 (a) A student carried out an investigation to determine the effect of different concentrations of a substrate on the activity of an enzyme. Another set of experiments was carried out under the same conditions but with a compound **X** added. The results for both experiments were sketched out by the student in Fig.1.1 below.

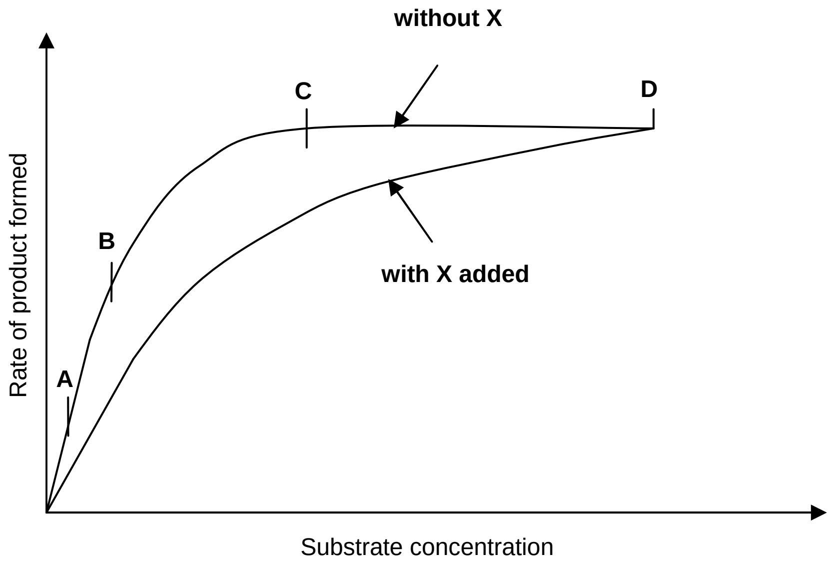

Fig.1.1

Account for the shape of the graph (without **X**)

(i) from **A** to **B**, and

- **The activity of the enzyme is <u>limited by substrate availability</u> (wtte)**
- **As substrate concentration increases, <u>increasing frequency of successful collisions</u> (hence formation of <u>enzyme-substrate complexes</u>)**
- **leading to formation of more product per unit time [max 2]**

(ii) from **C** to **D**.

- **Increasing substrate concentration does not increase the activity of the enzyme**
- **Enzyme active sites saturated by substrate molecules**
- **Concentration of <u>enzyme</u> is most probably <u>limiting</u> [max 2]**

(iii) Account for the effect of **X** on the enzyme.

- **Enzyme undergoes <u>competitive inhibition</u>**
- **X may be <u>structurally similar to substrate</u>**
- **X competes with substrate for the occupation of <u>active sites</u> of enzyme molecules, <u>reduced</u> level of product formed (reduced level of enzyme activity)**
- **At higher substrate concentrations of substrate, substrate molecules out-compete inhibitor molecules for enzyme sites, hence inhibition can be overcome [max 3]**

(b) When tadpoles turn into frogs, their tails become shorter and finally disappear. Fig.1.2 shows how the concentration of a protein-digesting enzyme, cathepsin, found in lysosomes changes as a tadpole matures into a frog.

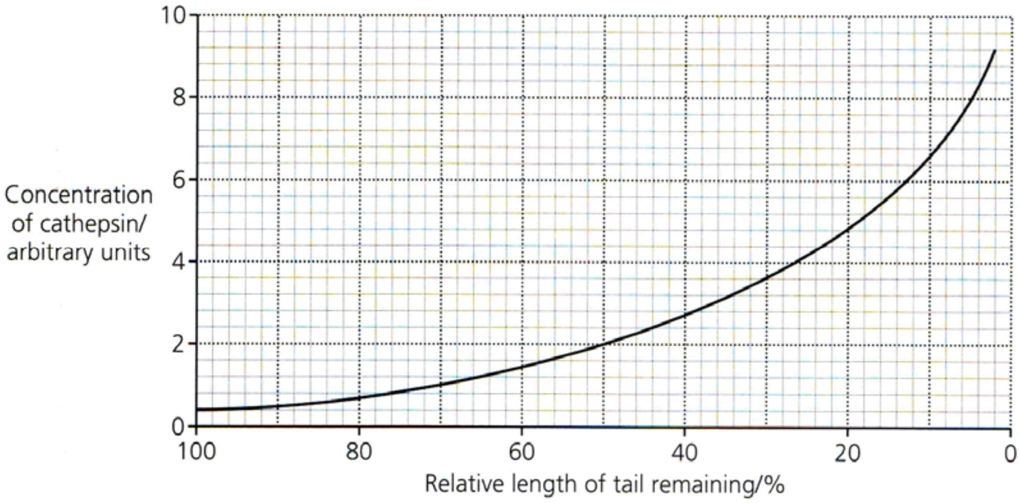

Fig.1.2

(i) Suggest two possible causes for the increase in cathepsin concentration in Fig.1.2.

- **Phagocytic cells containing cathepsin e.g. macrophages, increase in number during the regression of the tadpole tail, leading to increased levels of cathepsin**
- **Inactive enzymes (proenzymes) become activated**
- **Lysosomes in cells burst, releasing cathepsin**
- **Any Valid Points [max 2]**

(ii) Explain an advantage to the developing frog when the proteins in the tadpole's tail break down and the products are absorbed.

- **Reabsorbed products of protein breakdown (e.g. amino acids) are used to build up <u>new protein</u> molecules**
- **In limbs and organs etc. of developing tadpole, material from the breakdown of the tail is <u>not wasted</u>**
- **Developing tadpole need not have to acquire all the amino acids from diet [max 2]**

(iii) Apart from protein-digesting enzymes, suggest one other type of digestive enzyme that you would expect to show a similar change in concentration as the tadpole's tail shortens. Give a reason for your answer.

- **RNase (ribonucleases), Dnase (deoxyribonucleases), phospholipases, glycosidases etc. or any valid example**
- **Appropriate reason with reference to the type of molecules present e.g. phospholipases to breakdown structural phospholipids molecules in cell membranes of cells of the tail or Dnases to breakdown DNA in nucleus of unwanted cells etc or any valid point [2]**

[Total: 13]

[Turn over

2 **(a)** Fig.2.1 shows cells from an onion root tip. The root tip has been squashed and stained to show the stages of mitosis.

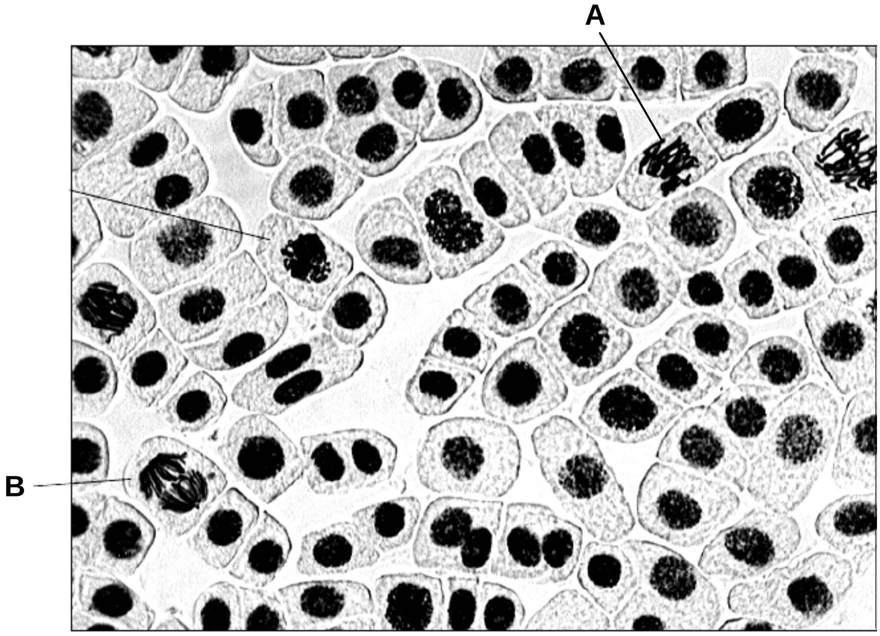

Fig.2.1

(i) Describe what happens to the chromosomes during stage **A**.

- **is <u>metaphase</u>**
- **<u>Chromosomes</u> align along the <u>equator</u>/<u>metaphase plate</u>**
- **Chromosomes attached to <u>spindle</u> at <u>centromeres</u> [max 2]**

(ii) Explain how stage **B** ensures that the daughter cells formed will be genetically identical.

- **<u>Genetically identical</u> <u>sister chromatids</u>**
- **<u>Centromere divides</u> and chromatids are <u>separated</u> to opposite poles**
- **<u>Shortening of spindle fibres</u> [max 2]**

(iii) The amount of DNA in cells from the onion root tip tissue was analysed. Some cells were found to have 7.6 units of DNA, others had only 3.8 units. Explain why this is so.

- **7.6 is replicated DNA / cell with 2 sister chromatids joined together at late interphase / prophase / metaphase/ anaphase / <u>early</u> telophase**
- **3.8 contains single chromatids / DNA is not replicated / <u>late</u> telophase / early interphase/ before S phase of interphase; non-dividing cells**

OR

- **Polyploidy due to non-disjunction of chromosomes during anaphase**
- **Results in an extra set of chromosomes in the cell (7.6 units) [2]**

(b) Fig.2.2 shows the chromosomes from a cell undergoing meiotic division.

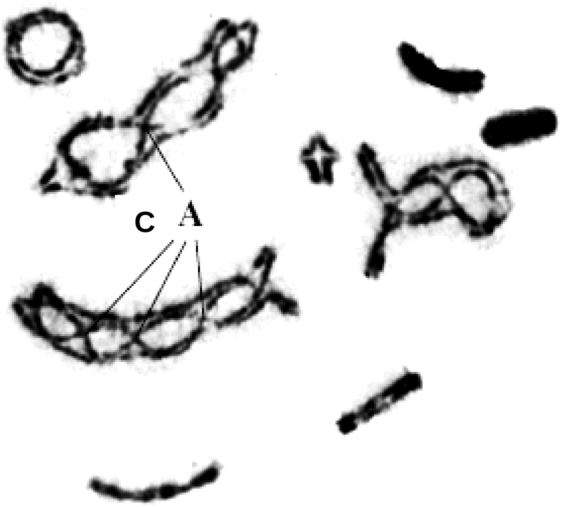

Fig.2.2

(i) Name the phase shown in the photograph. Give a reason for your answer.

- **Prophase I / Prophase of Meiosis I**
- **Pairing up of homologous chromosomes to form bivalents / Chiasmata present between homologous chromosomes / Bivalents are not yet aligned at the equator [2]**

(ii) What is the chromosome number of this cell at the end of meiosis II?

- **9 [1]**

(iii) Explain the significance of forming this number of chromosomes at the end of meiosis II.

- **Haploid chromosome number in the gametes allow the restoration of diploid chromosome number in the fertilized cell upon fusion of gametes [1]**

(iv) What has happened at the points labelled **C** in the photograph? Explain the significance of forming these points.

- **Points C are chiasmata formed between the non-sister chromatids of homologous chromosomes**
- **Allows crossing over / exchange of parts of the chromatids**
- **Leads to new linkage groups / recombinations of alleles in the chromosomes**
- **Gives rise to genetic variation**
- **Genetic variation in population allows natural selection to take place [max 3]**

[Total: 13]

3 (a) In both respiration and photosynthesis, ATP molecules are produced. However, there are significant differences between the 2 processes.

(i) Discuss 3 similarities in the production of ATP in respiration and photosynthesis.

- **flow of / passing electrons down electron transport chain**
- **generation of electrochemical / proton gradient across membrane**
- **diffusion (R: movement/flowing) of protons through stalk particles/ATP synthase or proton motive force to drive phosphorylation of ADP**
- **mobile electron acceptors such as NAD^+ and FAD in respiration vs $NADP^+$ in photosynthesis pass electrons**
- **chemical energy originated from substrates such as glucose in respiration vs light energy in photosynthesis [max 3]**

(ii) Discuss two structural adaptations of mitochondria for its function in ATP synthesis. [any 2]

- **Cristae / convoluted inner membrane**
  – increase surface area for embedment of e^- carriers / stalked particles etc ;
- **Presence of stalked particles/ ATPase etc**
  – for ATP synthesis
- **Inner membrane impermeable to H^+**
  - allows generation of H^+ gradient/accumulation of H^+/ proton motive force
- **Cytochromes arranged close to each other**
  – efficient electron transfer
- **Compartmentalisation**
  - separates substrates, enzymes and reaction conditions of Kreb's cycle/OP/ATP synthesis in mitochondria from glycolysis in cytosol [max 2]

(b) Explain the effect (if any) of a decrease in carbon dioxide concentration in

(i) respiration, and [1]

- **No effect - CO_2 is not substrate / not raw material / not required / product [all or nothing] [1]**

(ii) Photosynthesis [2]

- **decrease in photosynthesis / less glucose**
- **for carboxylation of RuBP in Calvin cycle / dark reaction / light independent reaction [2]**

(c) A suspension of mitochondria was isolated from liver tissue. Various substances were added to the suspension at different time intervals and the amount of oxygen remaining in the preparation was monitored over some time. Fig.3.1 shows the results as well as the times at which different substances were added.

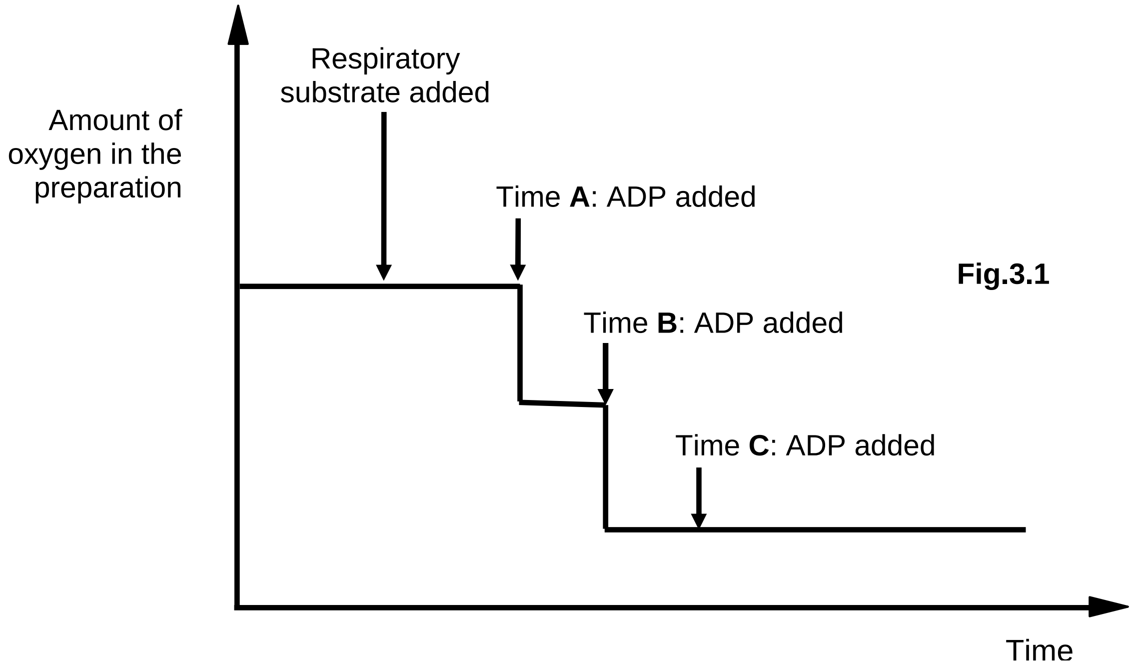

Fig.3.1

(i) Explain why glucose cannot be the respiratory substrate that was added.[2]

- **Lack of cytosol/cytoplasm in the suspension for glycolysis to produce pyruvate, or lack of enzymes involved in glycolysis to produce pyruvate**
- **Which is required as a substrate for Link reaction/Krebs cycle in mitochondria, or because glucose cannot be broken down/utilized by the mitochondria [2]**

(ii) Explain the change in the amount of oxygen between Time A and Time B.[3]

- **ADP is required as a substrate which is phosphorylated, synthesising ATP**
- **Hence when ADP is added at Time A, OP is allowed to proceed**
- **And more O_2 is used as the final electron acceptor [3]**

(iii) Account for the shape of the graph after Time C.[1]

- **Respiratory substrate / inorganic phosphates / NADH or FADH have been depleted [1]**

[Total: 14]

4 **(a)** Fig.4.1 shows two bacterial cells and their genetic material.

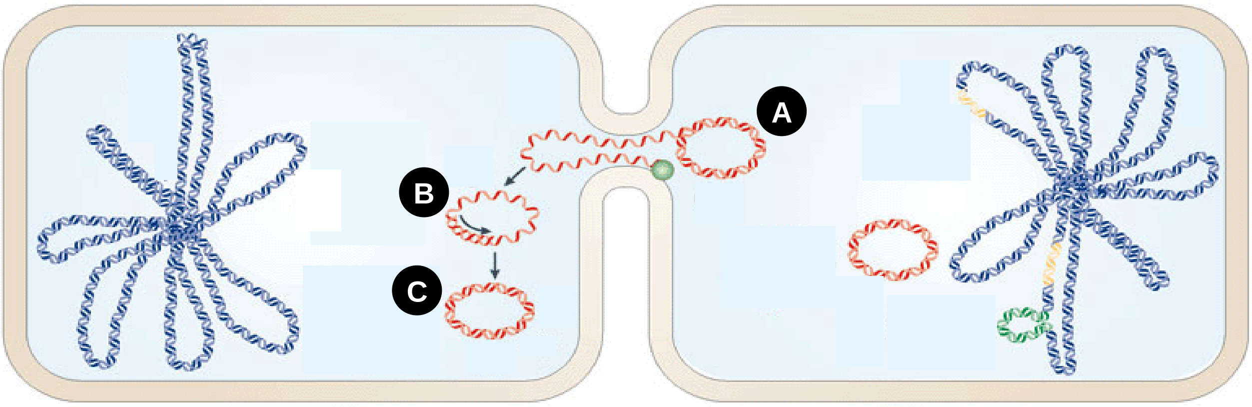

Fig.4.1

(i) State one other natural mechanism of genetic transfer between bacteria NOT shown in Fig.4.1.

- **transformation/transduction [1]**

(ii) Describe processes A to C.

- **<u>One</u> of the <u>plasmid DNA strands</u> is cut → a <u>single strand</u> is <u>transferred</u> across the <u>mating bridge</u> into the recepient cell [1]**
- **This strand now serves as a <u>template</u> for formation of the <u>complementary DNA strand</u> [1]**
- **A <u>double-stranded</u> plasmid DNA is formed [1]**

(iii) State one difference and one similarity between the bacterial chromosomal DNA, as seen in Fig.4.1, and a eukaryotic chromosome.

Difference:
no observed proteins associated with bacterial chromosome while histones are associated with eukaryotic chromosomes OR level of packing OR avp [1]

Similarity:
both exhibit packing/coiling/folding of DNA/double stranded DNA, avp [1]

(b) Fig.4.2 shows the process of RNA splicing, carried out by a complex of proteins and snRNA (small nuclear RNA) called a spliceosome. Splicing is one of the post-transcriptional modifications that occur in eukaryotic cells.

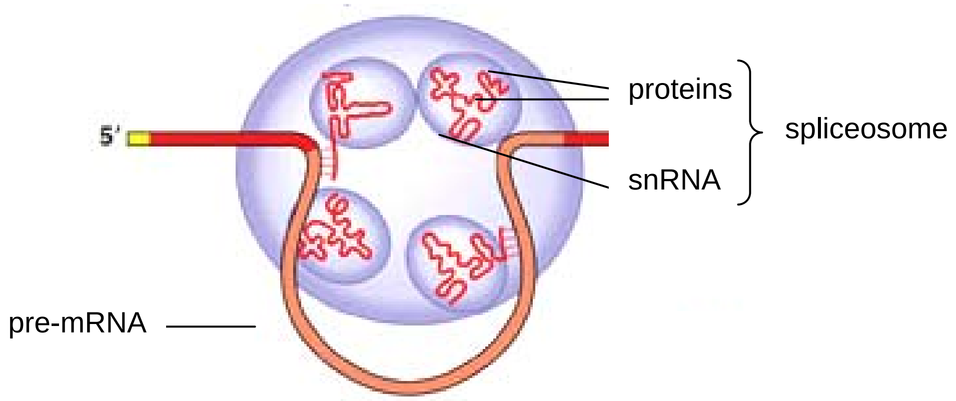

Fig.4.2

(i) Suggest a role for snRNA. [1]

- **Template for recognizing splice sites on the mRNA/specific sequences on mRNA where splicing should occur / brings the 2 splice sites together / allows enzymes to recognize splice sites [1]**

(ii) Describe how regulation at the post-transcriptional level can be carried out via alternative splicing. [3]

- **Spliceosomes recognise specific sequences at exon-intron interface**
- **Resulting in <u>excision</u> of different combinations of segments of introns (and exons) and certain exons in different mRNA**
- **Producing mature mRNA of different exons/sequences, hence giving different proteins [3]**

(c) Although environmental factors e.g. UV rays and diet, contribute to cancer, cancer is basically known as a genetic disease.

(i) Explain why cancer is basically known as a genetic disease. [2]

- **Cancer is caused by mutations in proto-oncogenes and tumour-suppressor genes,**
- **Resulting in abnormal proteins/protein levels that affect cell growth and division**
- **Environmental factors merely cause these gene mutations [max 2]**

(ii) Suggest how cancer at the **genetic level** is different from other genetic diseases such as cystic fibrosis and severe combined immunodeficiency (SCID)? [1]

- **CF and SCID are caused by single gene mutations**
- **cancer is a multi-step process involving a few gene mutations**

OR

- **multiple genes**

- **single genes**

OR

- **dominant/recessive**
- **recessive**

OR

- **may not be inherited**
- **inherited [any point, each 2 marks]**

[Total: 14]

5 (a) Fig.5.1 is a pedigree which traces the inheritance of alkaptonuria, a biochemical disorder. Affected individuals, indicated here by the coloured circles and squares, are unable to break down a substance called alkapton, which colours urine and stains body tissues.

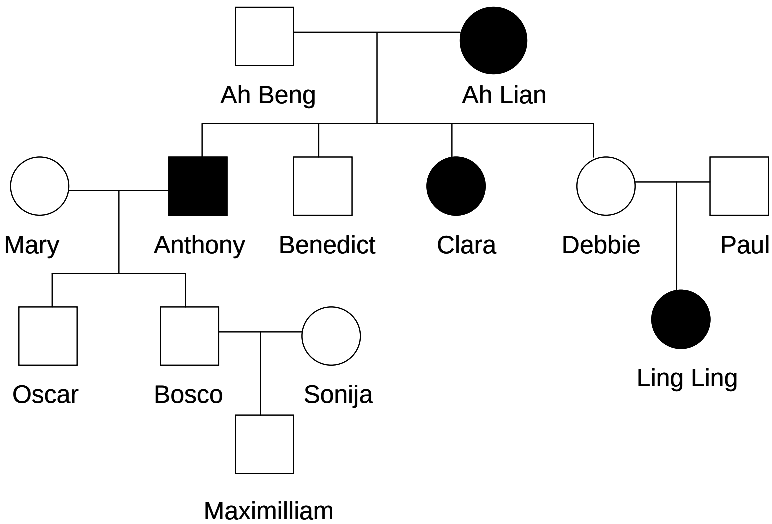

Fig.5.1

(i) Is alkaptonuria a dominant or recessive disorder? Explain your answer.

- **Recessive AND**
- **Benedict and Debbie are not affected by Ling Ling is affected/must be heterozygous for Ling Ling to be affected**
  [1 - no half mark awarded, all or nothing]

(ii) Name two individuals whose specific genotypes cannot be determined.

- **Mary, Sonija and Maximilliam [any 2]**

(iii) Oscar's genotype is Aa. If he marries a phenotypically normal female, what is the probability that his child will have alkaptonuria? Explain your answer.

- **Probability of phenotypically normal female cane be heterozygous/carrier, that is <u>Aa or AA is ½</u>**
- **Chances of having affected child with <u>heterozygous/carrier female</u> is ¼**
- **Therefore affected child with phenotypically normal female is <u>½ x ¼ = 1/8</u>**
  [3]

(b) An inbred variety of maize, **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

(i) Using appropriate symbols and indicating what they represent, state the genotype of **A**.
Let G=entirely green leaf and R=low resistance (or any appropriate symbols)
Genotype of A=ggrr [1]

The above F_1 generation was back-crossed to variety **A**. Using a genetic diagram, show the expected phenotype of the F_2 generation from this cross.

Parental cross (F_1 generation x variety A) **GgRr x ggrr**

(ii) **Gametes** **GR, gR, Gr, gr x gr**

1 mark

Punnett square (1 mark)

	GR	**gR**	**Gr**	**gr**
gr	**GgRr entirely green, low resistance**	**ggRr striped leaves, low resistance**	**Ggrr entirely green, resistant**	**ggrr striped leaves, resistant**

Expected phenotypic ratio (1 mark)	**1 entirely green, low resistance :**	**1 striped leaves, low resistance:**	**1 entirely green, resistant:**	**1 striped leaves, resistant**

[3]

The above was carried out (F_1 generation back-crossed to variety **A**) and yielded the following results:

finely striped leaves and resistance	80
finely striped leaves and low resistance	20
entirely green leaves and resistance	22
entirely green leaves and low resistance	78

The χ^2 (chi square) test was done on these data, giving a calculated value of χ^2 = 67.36.

	probability, p				
degree of freedom	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 5.1

(iii) Use the calculated value of χ^2 and the table of probabilities provided in Table 5.1 to find the probability of the results of the cross departing significantly by chance from the expected ratio.

Probability p <0.001

(iv) State what conclusions may be drawn from the probability found in **(b)(iii)**.

- **At a level of significance of 5%, since $p < 0.05$, we therefore reject the null hypothesis. (*not H^o → write out in full)***
- **The difference between observed and expected results is not due to chance alone.**
- **The predicted phenotype ratio of 1:1:1:1 is incorrect [max 2]**

Explain the discrepancy between the actual and expected results.

- **The loci/genes for leaf colour/pattern and resistance are linked/on same chromosome**
- **Accounts for larger *number of parental phenotypes* (78+92) finely striped & resistance + entirely green leaves & low resistance *{must specify parent phenotypes}***
- **Crossing over in meiosis 1 gives smaller number of recombinants phenotypes**
- **Smaller number recombinants phenotypes are finely striped leaves and low resistance + entirely green leaves & resistant *{must specify phenotypes}* [max 3]**

[Total: 16]

6 (a) Fig.6.1 shows how a stimulating electrode was used to change the potential difference across an axon membrane. Two other electrodes, **P** and **Q**, were used to record any potential difference produced after stimulation.

The experiment was repeated six times, using a different stimulus potential each time. In experiments **1** to **4**, the stimulating voltage made the inside of the axon less negative. In experiments **5** and **6**, it made the inside of the axon more negative.

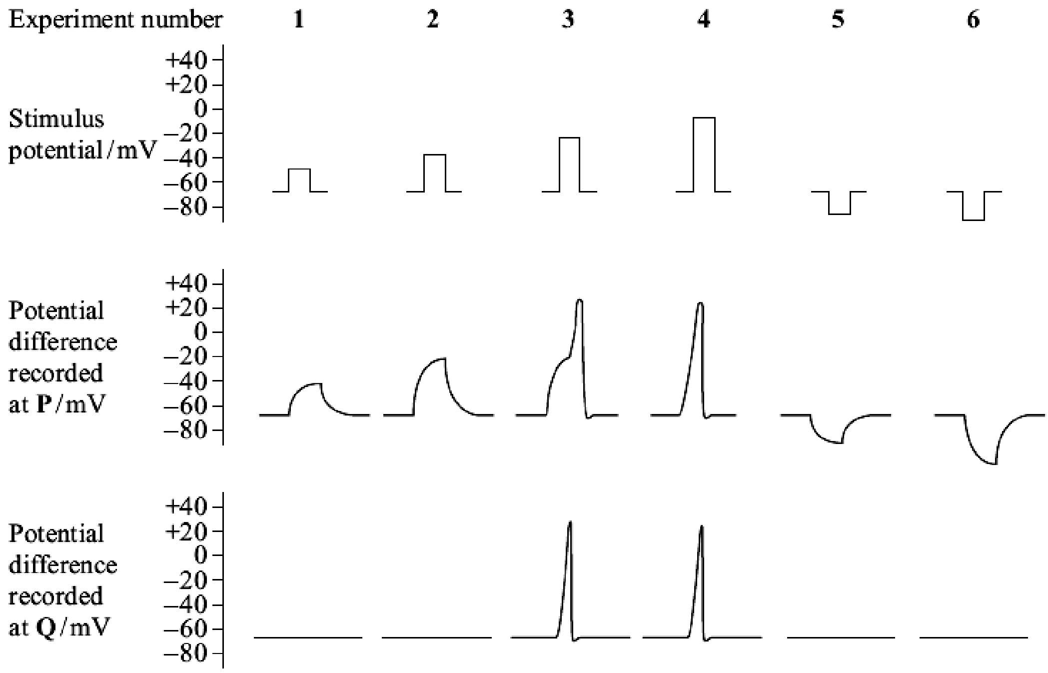

Fig.6.1

Explain the results of experiments **2** and **3**.

- **<u>Increasing stimulus</u> (potential) causes <u>decrease in potential difference / rise in potential at P/ less negative</u>; [Award if in either Expt 2 or 3]**

Expt 2

- **<u>Depolarisation</u> at P produces a <u>graded potential</u>**
- **<u>Below threshold potential</u> and <u>does not produce action potential</u>;**
- **Thus no impulse/action potential is conducted/propogated to Q [max 2]**

Expt 3

<u>Depolarisation</u> at P <u>exceeds threshold potential</u> and produces an <u>action potential</u>/depolarization from <u>-70mV to +30mV</u>

- **<u>Opening of Na</u> ion <u>voltage-gated channels</u> in axon membrane with <u>influx of Na+ ions</u>**
- **resulting in <u>impulse propagated (from P to Q)</u>;**
- **Action potential is <u>all-or-nothing</u> with <u>no change in amplitude</u> of peak [max 2]**

(b) Myasthenia gravis is a disease which causes muscular weakness. It develops because of an attack by the body's own immune system on neuromuscular junctions. Fig.6.2 below shows a normal neuromuscular junction (nerve to muscle) and one affected by the disease (myasthenic).

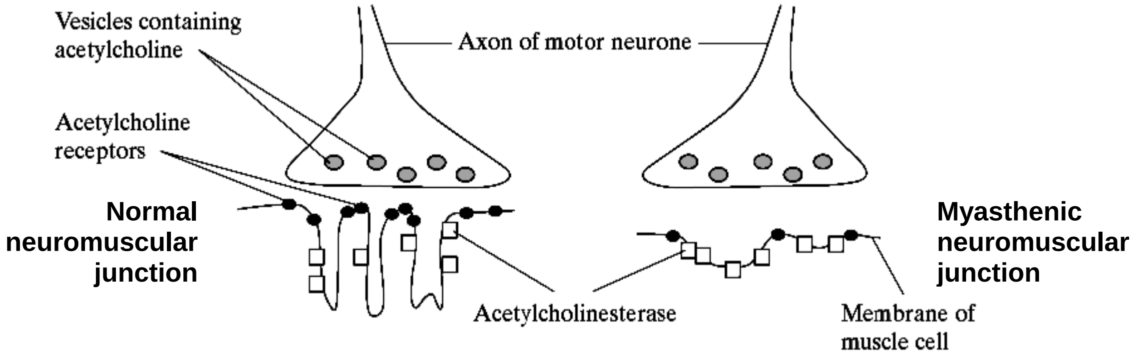

Fig.6.2

Describe two ways in which a myasthenic neuromuscular junction differs from a normal one and explain how each difference would affect transmissions across the myasthenic neuromuscular junction.

1. **Difference: myasthenic has <u>fewer folds/ less surface area</u> thus <u>fewer receptors</u>;**
   Effect: so <u>less</u> chance of <u>depolarization</u>(accept excitatory postsynaptic potential) /fewer <u>Na+ ligand gated channels open</u> for Na+ influx;

2. **Difference: myasthenic has acetylcholinesterase in <u>shallower folds/more exposed</u>; Or**
   myasthenic has lower ratio of receptors to acetylcholinesterase;
   Effect: so acetylcholine are <u>degraded/destroyed/hydrolysed before binding to acetylcholine receptors</u> or mention <u>faster degradation</u> of transmitter

3. **Difference: myasthenic has <u>wider gap/cleft</u>;**
   Effect: so <u>takes longer</u> for neurotransmitters to <u>diffuse across</u>;
   [any 2 differences and their corresponding effects]

(c) Multiple sclerosis is a disease in which the myelin sheath surrounding neurones are destroyed and so the neurones become de-myelinated. People with multiple sclerosis produce slow responses to stimuli.

Explain how myelination of neurones help in speeding up responses.

- **Myelin sheath contains fatty substances/lipids acts as an <u>insulator against depolarization</u>/ Na+ and K+ ions cannot pass across it/ AP cannot produce**
- **Local circuit produced by action potential in one node travels to the next node, causes another action potential to be produced**
- **So that action potential can <u>jump</u> from <u>one node to node/ saltatory conduction</u> [max 2]**

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

7 **(a)** Discuss the two categories of transport proteins found in cell membranes.

transport proteins have hydrophilic domains that allow hydrophilic/charged,polar
molecules that cannot transverse the hydrophobic core of the phospholipids bilayer to pass through;
category 1 : channel proteins - are transmembrane proteins;
may be gated, which means they can open to allow the facilitated diffusion of solutes or they can be closed to prohibit diffusion;
this feature allows cells to regulate the movement of solutes. In addition, solutes move
through channels from a high to a low solute concentration, down a concentration gradient;
e.g.voltage gated Na^+ channel;
may not be gated form an open passageway for the direct facilitated diffusion of ions or molecules across the membrane;
e.g. aquaporins;
category 2: carrier proteins;
bind the solute on one side of the membrane which results the protein undergoing a conformational change that allows access of the solute to the opposite side of the membrane;
pumps are a type of protein carrier that usually use ATP to move solutes against a
concentration gradient (from a low solute concentration to a high solute concentration).
e.g. Na^+ - K^+ pump;
some carriers transport solutes by facilitated diffusion,down a concentration gradient
e.g. glucose transport carriers, Glu T1, across red blood cells

(b) Describe the role of mRNA in protein synthesis. [6]

1. **mRNA is synthesized in the nucleus by transcription via complementary base pairing**
2. **transcription is done via complementary base pairing to ensure that the information in DNA is accurately transferred for translation**
3. **mRNA, being a single-stranded molecule is small enough to**
4. **move out of nucleus via nuclear pores into cytoplasm**
5. **where it attaches to ribosome to start translation**
6. **mRNA that moved out to cytoplasm consist of a coding segment which is made up of triplet codes / codon, one codon per amino acid, which determines the linear sequence of amino acid**
7. **the start codon or AUG on the mRNA initiates translation coding for methionine**
8. **and the stop codon (UAG, UAA, UGA) terminates translation**
9. **codons on mRNA matches, via complementary base pairing, the anti-codon in tRNA which carries a specific amino acid**
10. **a single mRNA can be attached by many ribosomes (polysomes) to synthesise multiple copies of the same polypeptide increasing speed of translation**
11. **mRNA molecules can be spliced in different ways to produce different protein / polypeptide**

(c) Explain with examples, how anatomical and molecular homologies support Darwin's theory of evolution. [8]

Define anatomical homology
-Anatomical homologies are structures such as bone, organs and any other physical structures in an organism that are derived from a <u>common ancestor</u>; when comparing between organisms; [1]

-Examples of anatomical homologies are the pentadactyl limb, veterbrate column, presence of hair, mammary glands found in animals that share these homologies; [1]

Define Molecular homologies
-Organisms with <u>molecular homology</u> have similar genes & amino acid sequences that are derived from a common ancestor. [1]

-Molecular homology also refers to biochemistry that are derived from a common ancestor as well [1]

-Examples of homologous genes in different animals are the haemoglobin, ribosomal, cytochrome oxidase genes. These genes are derived from a common ancestor.[1]

How it supports Darwin's theory of evolution

<u>-Homologies suggest common ancestry</u>: e.g. The pentadactyl limb is a homologous feature of all tetrapods, with different forms in different species. This suggests that they descended from a common ancestor which had a basic form of the pentadactyl limb. Like wise for molecular homology. [1]

<u>-Hence, homologies show "descent with modification"</u>: Comparisons of homologies (anatomical, embryological and molecular) between species show how an ancestral homology in a population may have been modified in descendent species through natural selection. [1]

<u>-Homologies provide the basis of comparison:</u> hence homologous traits are what are compared between populations or species as they are derived from a common ancestor which shows the modification process from a basic ancestral form. For molecular homology, to compare two species one must look at homolgous genes/aa sequence. [1]

<u>-Non-homologous (analogous) structures,</u> however, (such as the wing of a bird and the wing of insects) do not provide such a basis as they are derived from different ancestors, although they may have a similar function, [1]

8 (a) Discuss how regulation of gene expression can occur at the translational and post-translational level in a eukaryotic cell. [8]

At the translational level of regulation,

1. regulatory proteins;
block translation by preventing ribosome attachment;
2. elongate poly(A) tail which was originally shortened;
to signal for initiation of translation;
3. phosphorylation/dephosphorylation of translation initiation factors;
inhibit translation by binding to factor/activate by releasing factor or e.g eIF2 and eIF4;
4. length of poly(A);
because enzymes will degrade the tail once mRNA enters the cytosol, and once the tail reaches a certain length, the mRNA will be degraded rapidly;
(same point for certain UTR regions having greater affinity for ribosomes and presence of 5' cap protectin mRNA from degradation)
5. presence of RNAi e.g. miRNA/siRNA;
binds to complementary segments on mRNA, hence preventing translation/forms a complex with an enzyme which will degrade any mRNA which has complementary segments to the siRNA;

At the post-translational level of regulation,

6. phosphorylation/glycosylation etc (any biochemical modification that sets the protein for a role in the cell);
activate(or deactivate) the protein/ targets the protein for association with membranes etc (any accurate role of the modification);
7. attachment of ubiquitin;
marks the protein for degradation at a proteosome;

(b) Discuss how signal amplification is illustrated by the effect of hormone on glycogenolysis. [8]

1. Binding of **one / a molecule of epinephrine / glucagon** binds to its GPCR to form **ligand-receptor complex ;;** will cause
2. **GPRC** will undergo a **conformation change ;;**
3. to allow the **activation of several / few G-protein** by the **losing of GDP for binding GTP ;;**
4. Each activated G-protein will translocate and to **activate** enzyme **adenylyl cyclase,** each of which is able to catalyse the conversion of **large number of ATP to cAMP ;;** hence **amplifying** the signal.

   [R: increase cAMP, activation of cAMP]
5. These cAMP in turn **binds** and **activates** to **large number of protein kinase A ;;**.

   [R:cAMP phosphorylate kinases]
6. Each activated protein kinase A will initiate a **sequential phosphorylation** and **activation of kinases / phosphorylation cascade ;;**
7. At each phosphorylation step/cascade, **each activated kinase is able to activate a large number** of the next relay molecule/kinase ;; hence the **signal is further amplified**.

8a. lead to the activation of **large number of glycogen phosphorylase**

8b. each will catalyse for the breakdown of **large amount** of **glycogen into glucose**.

[R: syn large amount of glucose]

9. the **number of activated product** is **always greater than those in the preceding step** as one move down the cascade **;;**
10. binding of **1 glucagon** to receptors will lead to the hydrolysis of **large number of glycogen;;**

** only if amplification is clearly illustrated for each of the points numbered 3-8, a that marking point will be marked with a circle. If none of the marking points 3-8 are not circled, 1 mark will be deducted from the overall marks.

(c) Briefly describe with named examples, the effect of extracellular ligand in cell signalling in [4]

(i) endocrine system.

1. **<u>Glucagon</u> binds to cell surface <u>receptors/GPCR</u> of <u>muscle / liver</u> cells ;;**
2. **stimulating the conversion of glucose to glycogen ;;**

<u>OR</u>

1. **<u>Insulin</u> binds to cell surface <u>receptors / RTK</u> of <u>muscle / liver</u> cells ;;**
2. **stimulating the conversion of glycogen to glucose / increase permeability to glucose / glucose uptake ;;**

(ii) nervous system.

11. **<u>Acetylcholine</u> binds to cell surface <u>receptors</u> of <u>post-synaptic knob / membrane</u> ;;**
12. **increasing permeability to Na+/K+ / leading to depolarization / hyperpolarisation / EPSP / IPSP ;;**

[Total: 20]